

# QIN YANG

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Power BI Data Analyst Associate | Snowflake SnowPro Core | Microsoft Azure Data Scientist Associate

## SUMMARY

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Proven track record in building end-to-end ML pipelines, SQL-based data engineering, and translating complex data into actionable business insights. Experienced in deploying production-grade models, creating enterprise-level dashboards for stakeholder communication, and collaborating cross-functionally to support data-driven decision-making.

## EXPERIENCE

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**Data Scientist, Marketing** @Tractor Supply Co.

*Brentwood, TN | 2025 Jan - Present*

- **Data Engineering & Pipelines:** Engineered scalable SQL pipelines integrating 1M+ daily records (site traffic, financial) into Snowflake, creating analytically-ready datasets for downstream modeling and reporting.
- **Modeling & Causal Inference:** Designed and executed A/B tests with rigorous statistical analysis to quantify marketing campaigns effectiveness; translated results into clear recommendations for cross-functional stakeholders including Marketing, Finance, and Merchandise teams.
- **Recommendation Systems:** Building purchase propensity models using XGBoost on Databricks, with MLflow for experiment tracking and model versioning; outputs feed downstream recommendation systems to identify high-intent customer segments and drive personalized marketing strategies.
- **Dashboards & Visualization:** Built automated Power BI dashboards using Snowflake SQL and Git-based workflows, enabling real-time monitoring of KPIs and replacing manual reporting processes.
- **Stakeholder Communications:** Partnered with VP-level leadership to define success metrics and standardize reporting frameworks, ensuring data outputs directly informed budget allocation decisions.

**Data Scientist, Global Automation & Engineering** @Assurant

*Nashville, TN | 2023 Feb - Jan 2025*

- **ML Pipelines & MLOps:** Developed end-to-end ML models and optimized CI/CD pipelines using Docker, Azure ML, MLflow, scikit-learn, and Git within shared codebase - directly supporting reproducible, production-ready workflows.
- **Applied ML (NLP + Computer Vision):** Implemented a combined NLP + Computer Vision algorithm to automate a mobile device workflow, reducing labor costs by \$300K annually and creating a new external revenue stream.
- **Real-Time Data Visualization Apps:** Led development of real-time Streamlit/Django data visualization apps integrating databases, APIs, and IoT data sources, enabling warehouse operators and executives to access critical insights quickly.
- **Data Analysis & Reporting:** Used Python, Pandas/Numpy, SQL, and PySpark to conduct in-depth analysis on large-scale datasets; built dynamic visualizations that translated complex outputs into actionable recommendations for VP-level leadership.
- **Collaboration & Communication:** Facilitated cross-functional brainstorming sessions and tracked project progress with JIRA, ensuring timely delivery of data initiatives aligned with organization goals.

**Research Assistant (Automatic Speech Recognition, ASR),** Center for Data Science@NYU

*New York, NY | 2022 Sep - 2023 May*

- **NLP & Model Development:** Fine-tuned an Automatic Speech Recognition model on 375 hours of labeled and pseudolabeled audio data using Pytorch; integrated a word-based language model, reducing word error rate to 13.5%.
- **Reproducible ML Workflows:** Tracked experiments using Weights and Biases and High-Performance Computing (HPC) to ensure reproducible workflows and accelerated training.
- **Research & Academic Writing:** Co-authored and published research at the *2023 Interspeech Conference*, demonstrating strong technical writing and communication skills.

**Data Science Intern, Digital Transformation**@Campana & Schott

*New York, NY | 2022 May - 2022 Sep*

- **ETL & Data Pipelines:** Built robust ETL pipelines to extract data from APIs and databases, transform/clean it, and load into centralized repositories for modeling and analysis.
- **Applied NLP & Time Series:** Applied NLP techniques (NER, Sentiment Analysis) and time-series models (LSTM, N-HiTs) to extract insights from unstructured pharmaceutical data, supporting strategic planning.
- **Power BI Dashboards:** Developed Power BI dashboards to visualize KPIs and trends, ensuring stakeholders to monitor business metrics and make data-informed decisions.

## EDUCATION

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**Master's Degree in Computer Science (Online)** @Georgia Institute of Technology

*2025 Sep - Present*

Relevant Courses: Machine Learning for Trading, Database System Concept and Design, Data and Visual Analytics

**Master's Degree in Data Science** @New York University

*2021 Sep - 2023 May*

Relevant Courses: Machine Learning, Applied ML in Finance, NLP and Computational Semantics, Deep Learning, Time Series Analysis, Probability & Statistics, Big Data

**Master's Degree in Economics / Bachelor's in Management Science & Engineering** @Shandong Univesity

*2014 Sep - 2021 May*

Relevant Courses: Advanced Micro & Macro Econ, Econometrics, Finance, Calculus, Linear Algebra, Data Principles and Applications, Python and Data Analysis, Database Management

## SKILLS & TOOLS

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**Programming Languages:** Python, SQL

**Machine Learning:** Regression, Classification, & Clustering, Recommendation Systems (e.g., RNN-based, DNN-based, Popularity-based, Collaborative Filtering, etc.)

**AI:** NLP (Text Classification, Sentiment Analysis, NER, Chatbot, RAG), ASR

**Visualization:** Power BI, Plotly, Dash

**DBs:** PostgreSQL, MongoDB, MySQL, Vector DB, Azure Cosmos DB

**Other:** Time Series Analysis, Langchain, Linux, Multithreading, Multiprocessing, Office, HTML, CSS, Git, FastAPI, PyTorch, Tensorflow, Mlflow, Flask, Django, Azure ML, Streamlit, Docker, Data Analytics, RESTful API, CI/CD, Version Control, ETL, JIRA, Snowflake, Adobe Analytics, Databricks, PySpark

**Soft skills:** Stakeholder Engagement, Interdisciplinary Communication, Collaborative Problem Solving, Project Management, Feedback & Iteration, Cross-team Collaboration, Advanced Analytics, Supply Chain, Marketing Analytics